

BACHELORS OF SCIENCE IN SOFTWARE ENGINEERING-BS(SE)

Structure of BS(SE) Program

Category: Courses (Credit Hours)

Total Credit Hours: 134 Total Courses: 47 Semesters: 8 Duration: 4 Years	General Education: 15 (34) Allied/interdisciplinary courses: 05 (15) Major Courses : 18(58) + 7(21) Internship/field experience: Qualifying Total: 47 (134)
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S#	COURSE CODES	CLASS	COURSE TITLE	CREDIT HOURS	PRE-REQ		CO-REQ
SEMESTER 1							
1	CMC111	CORE	Programming Fundamentals	3 + 0	-		CMC111-L
	CMC111-L	CORE	Programming Fundamentals (Lab)	0 + 1	-		
2	GER111	GER	Application of Information & Communication Technologies	2 + 0	-		GER111-L
	GER111-L	GER	Application of Information & Communication Technologies (Lab)	0 + 1	-		
3	GER121	GER	Functional English	3 + 0	-		
4	GER131	GER	Calculus and Analytic Geometry	3 + 0	-		
5	GER152	GER	Natural Science (Applied Physics)	2 + 0	-		GER151-L
	GER152-L	GER	Natural Science (Applied Physics) (Lab)	0 + 1	-		
				13+3(16)			
SEMESTER 2							
6	MTE111	MTE	Multivariable Calculus	3 + 0	GER131		
7	CMC112	CORE	Object Oriented Programming	3 + 0	CMC111		CMC112-L
	CMC112-L	CORE	Object Oriented Programming (Lab)	0 + 1	CMC111		
8	CMC123	CORE	Digital Logic Design	2 + 0	GER152		CMC121-L
	CMC123-L	CORE	Digital Logic Design (Lab)	0 + 1	GER152		
9	GER122	GER	Expository Writing	3 + 0	GER121		
10	GER241	GER	Pakistan Studies	2 + 0	MTE221		
11	GER132	GER	Discrete Structures	3 + 0	GER131		
12	GEN111*	GER	Understanding of Holy Quran- I*	0+1			
				16+3(19)			
SEMESTER 3							
13	MTE212	MTE	Probability & Statistics	3 + 0	MTE111		
14	CMC223	CORE	Computer Organization & Assembly Language	2 + 0	CMC123		CMC223-L
	CMC223-L	CORE	Computer Organization & Assembly Language (Lab)	0 + 1	CMC123		
15	CMC251	CORE	Data Structures	3 + 0	CMC112		CMC251-L
	CMC251-L	CORE	Data Structures (Lab)	0 + 1	CMC112		
16	CMC262	CORE	Computer Networks	2 + 0	CMC123#/GER 111		CMC262-L
	CMC262-L	CORE	Computer Networks (Lab)	0+ 1	CMC123#/GER 111		

17	SEC211	CORE	Introduction to Software Engineering	3 + 0	CMC112#/GER111		
18	GEN112*	GER	Understanding of Holy Quran- II*	0+1	GEN111*		
				13+4 (17)			
SEMESTER 4							
19	MTE213	MTE	Linear Algebra	3 + 0	MTE111		
20	GER221	MTE	Technical & Business Writing	3 + 0	GER122		
21	CMC242	CORE	Operating Systems	2 + 0	CMC251		CMC242-L
	CMC242-L	CORE	Operating Systems (Lab)	0 + 1	CMC251		
22	CMC252	CORE	Analysis of Algorithms	3 + 0	CMC251		
23	GER142	GER	Ideology and Constitution of Pakistan	2 + 0	GER142		
24	SEC231	DOMAIN CORE	Software Requirement Engineering	3 + 0	SEC211		
				16+1(17)			
SEMESTER 5							
25	SEC321	DOMAIN CORE	Software Quality Engineering	3 + 0	SEC231		
26	CMC363	CORE	Information Security	2 + 0	CMC262		CMC363-L
	CMC363-L	CORE	Information Security (Lab)	0 + 1	CMC262		
27	CMC331	CORE	Database Systems	3 + 0	CMC242		CMC331-L
	CMC331-L	CORE	Database Systems (Lab)	0 + 1	CMC242		
28	GERXXX	GER	Social Science I	2 + 0	GER122#/SEC211		
29	XXXXXX **	DOMAIN ELEC	Domain Elective 1	3**	--		
30	SEC311	DOMAIN CORE	Software Design & Architecture	3 + 0	SEC231		
				16+2(18)			
SEMESTER 6							
31	SEC322	DOMAIN CORE	Software Project Management	2 + 0	SEC211		SEC322-L
	SEC322-L	DOMAIN CORE	Software Project Management (Lab)	0 + 1	SEC211-L		
32	SEC312	DOMAIN CORE	Software Construction & Development	2 + 0	SEC311		SEC312-L
	SEC312-L	DOMAIN CORE	Software Construction & Development (Lab)	0 + 1	-		
33	XXXXXX **	DOMAIN ELEC	Domain Elective 2	3**	-		SEEXXX-L
34	CMC383	CORE	Artificial Intelligence	2 + 0	CMC252		CMC383-L
	CMC383-L	CORE	Artificial Intelligence (Lab)	0 + 1	CMC252		
35	GER141	GER	Islamic Studies	2 + 0	-		
36	GERXXX	ESC	Social Science II	3 + 0	SEC311/SEC331		
				13+4(17)			
SEMESTER 7							
37	CMC491	CORE	Final Year Project - I	0 + 3	CMC381 / SEC322 / CMC331		
38	XXXXXX **	DOMAIN ELEC	Domain Elective 3	3**	-		SEEXXX-L
39	XXXXXX **	DOMAIN ELEC	Domain Elective 4	3**	-		SEEXXX-L
40	XXXXXX **	DOMAIN ELEC	Domain Elective 5	3**	-		SEEXXX-L
41	CMC443	DOMAIN CORE	Parallel & Distributed Computing	2 + 0	CMC381#/CMC241		CSC443-L

	CMC443-L	DOMAIN CORE	Parallel & Distributed Computing (Lab)	0 + 1	CMC381#/CMC241		
42	GERXXX	GER	Entrepreneurship	2 + 0	SEC322		
				10+7(17)			
SEMESTER 8							
43	CMC492	CORE	Final Year Project - II	0 + 3	CMC491		
44	GER463	GER	Professional Practices	2 + 0	SEC322		
45	GER443	GER	Civics and Community Engagement	2 + 0	SEC322		
46	XXXXXX **	DOMAIN ELEC	Domain Elective 6	3**	-		SEEXXX-L
47	XXXXXX **	DOMAIN ELEC	Domain Elective 7	3**	-		SEEXXX-L
				8+5(13)			
			TOTAL CREDIT HOURS	134			

*Campus will decide to offer a 2 credit hrs. course for non-Muslim students.

** The offering of the 3+0 or 2+1 electives will depend on the availability of the teaching and lab resources

	Change in credit hrs.
	New course added
	Move to align with other programs and Fest policy
	Name Change

List of Domain Elective Courses for BS Software Engineering Program

List of Domain Elective Course(s) (3+0)			
S.No	Course Code	Title	Cr.Hrs
1	SEE311	Software Reverse Engineering	3 + 0
2	SEE341	Agent System Development	3 + 0
4	SEE321	Management Information System	3 + 0
5	SEE322	Agile Software Management	3 + 0
7	SEE314	Software Metrics	3 + 0
8	SEE315	Software Engineering Economics	3 + 0

List of Domain Elective Course(s) (2+1)			
S. No	Course Code	Title	Cr. Hrs
1	CSC382	HCI and Computer Graphics + Lab	2 + 1
2	CYE428	Blockchain Technologies + Lab	2 + 1
3	CSE477	DevOps + Lab	2 + 1
4	CSE317	Web Technologies + Lab	2 + 1
5	CSE384	Data Science + Lab	2 + 1
6	CSE383	Machine Learning + Lab	2 + 1
7	MGE454	Virtual and Augmented Reality + Lab	2 + 1
8	CSE416	Mobile Application & Development + Lab	2 + 1
9	CYC211	Cyber Security + Lab	2 + 1
10	MGC223	Game Design and Development + Lab	2 + 1
11	AIE423	Computer Vision + Lab	2 + 1
12	CSE474	E – Commerce + Lab	2 + 1
13	AIC322	Artificial Neural Networks & Deep Learning + Lab	2 + 1
14	AIE441	Natural Language Processing + Lab	2 + 1
15	CSE385	Data Warehousing and Data Mining + Lab	2 + 1
16	CYE312	Ethical Hacking and System Defense + Lab	2 + 1
17	SEE441	Full Stack Development + Lab	2 + 1
18	SEE451	High Performance Computing + Lab	2 + 1
19	SEE413	Software Verification and Validation (Testing & QA) + Lab	2 + 1

Social Science Electives				
S. No.	Course Code		Title	Cr. Hr
1	GER262	ESC	Sociology	2 + 0
2	GER261	ESC	Introduction to Management	2 + 0
4	ESC311	ESC	Introduction to Marketing	3 + 0
5	ESC212	ESC	Phycology	2 + 0
6	ESC312	ESC	Behavioral Sciences	3 + 0
7	GER263	ESC	Economics	3 + 0
8	GER361	ESC	Project Management	2 + 0